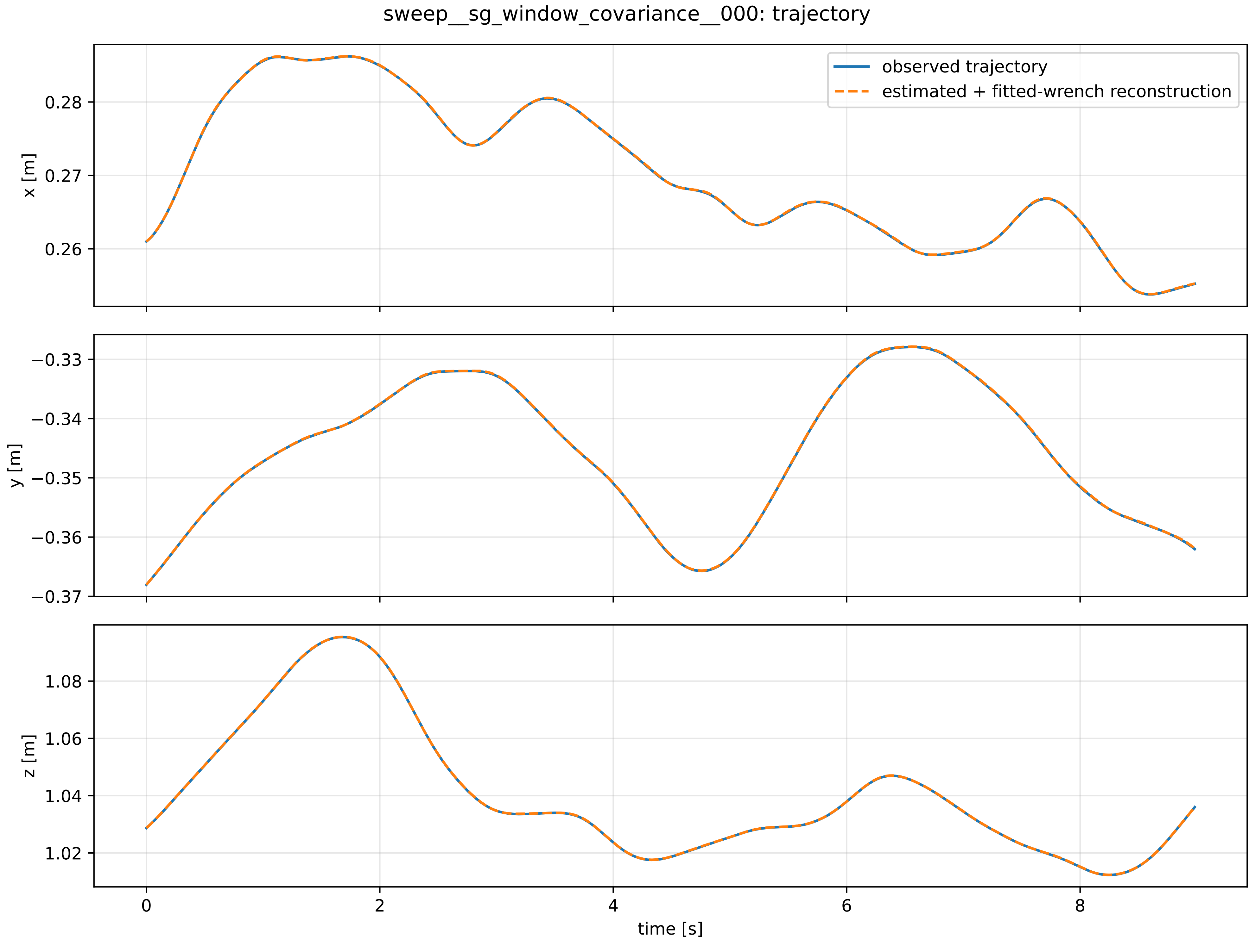
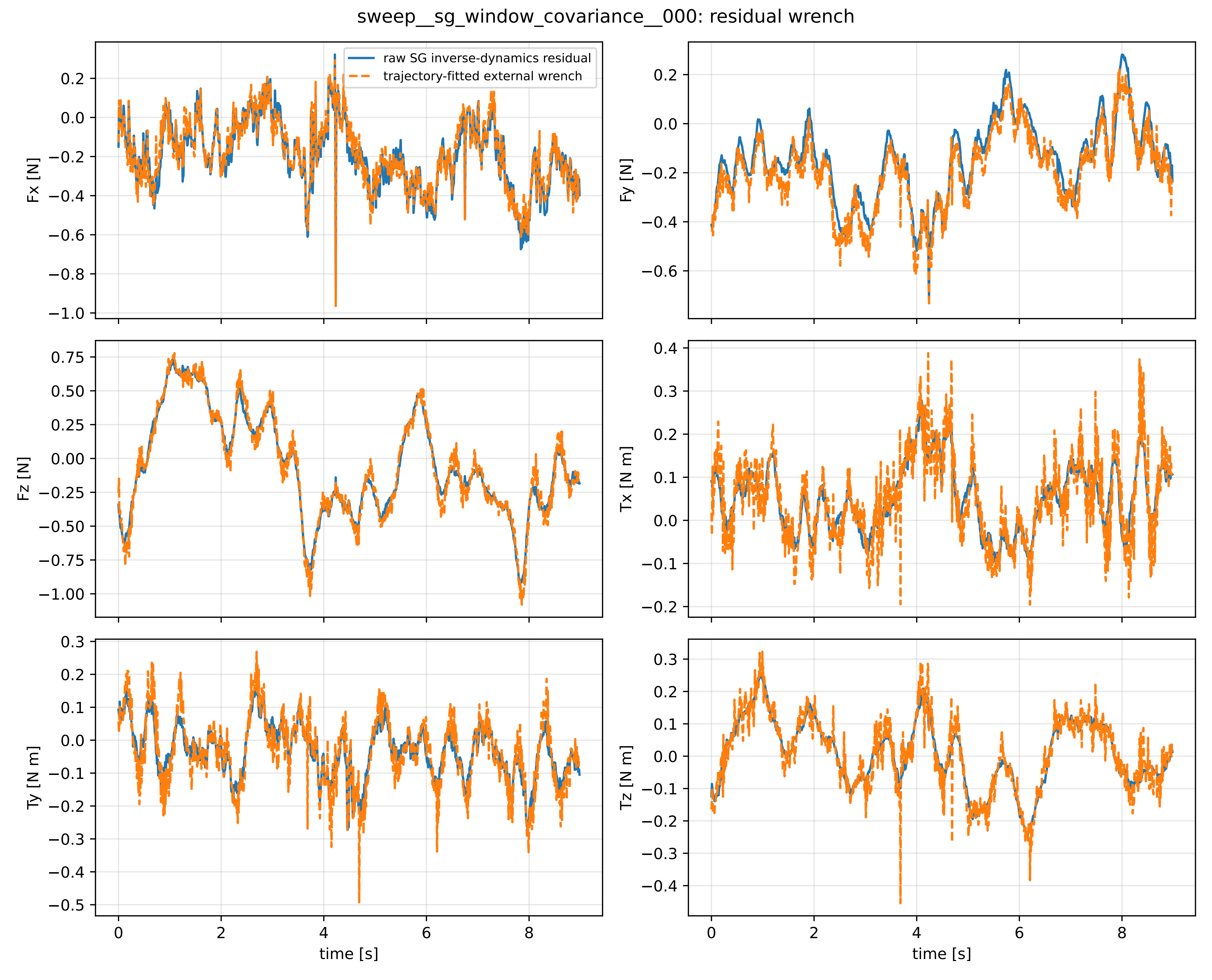
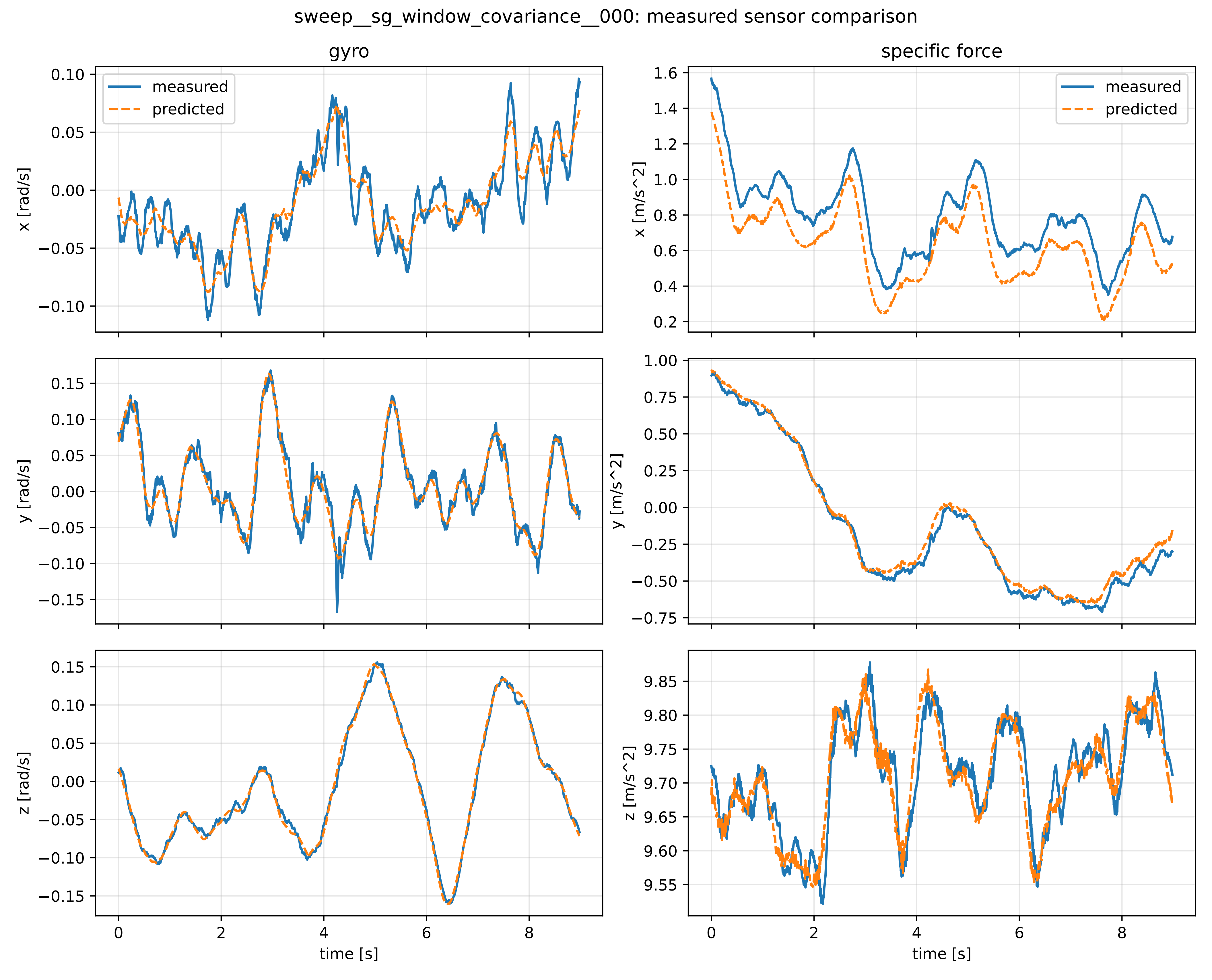

sweep__sg_window_covariance__000: trajectory
observed trajectory
estimated + fitted-wrench reconstruction
x [m]
0.28
0.27
0.26
y [m]
−0.33
−0.34
−0.35
−0.36
−0.37
z [m]
1.08
1.06
1.04
1.02
0
2
4
6
8
time [s]

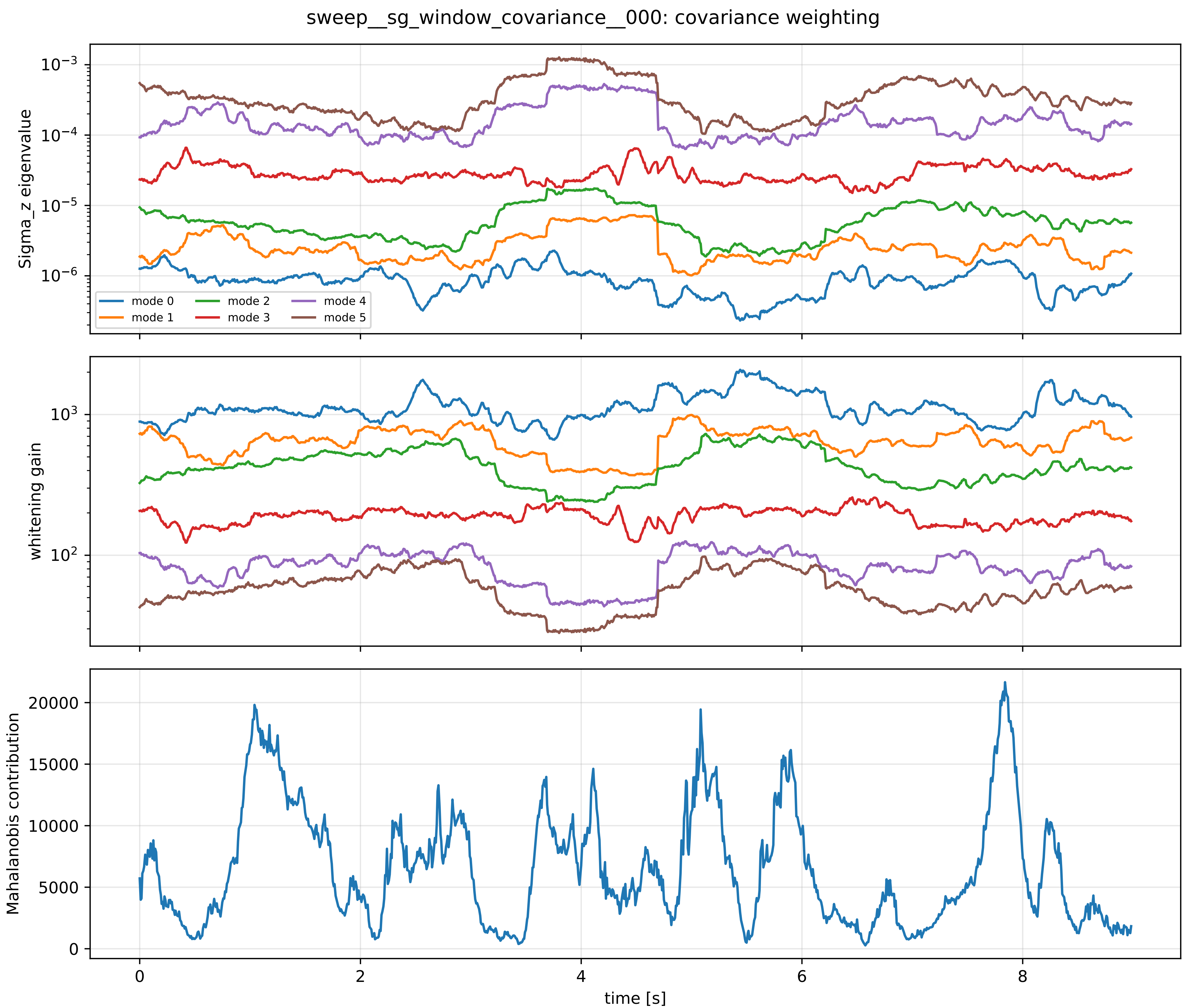


sweep__sg_window_covariance__000: residual wrench
raw SG inverse-dynamics residual
trajectory-fitted external wrench
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
time [s]
time [s]

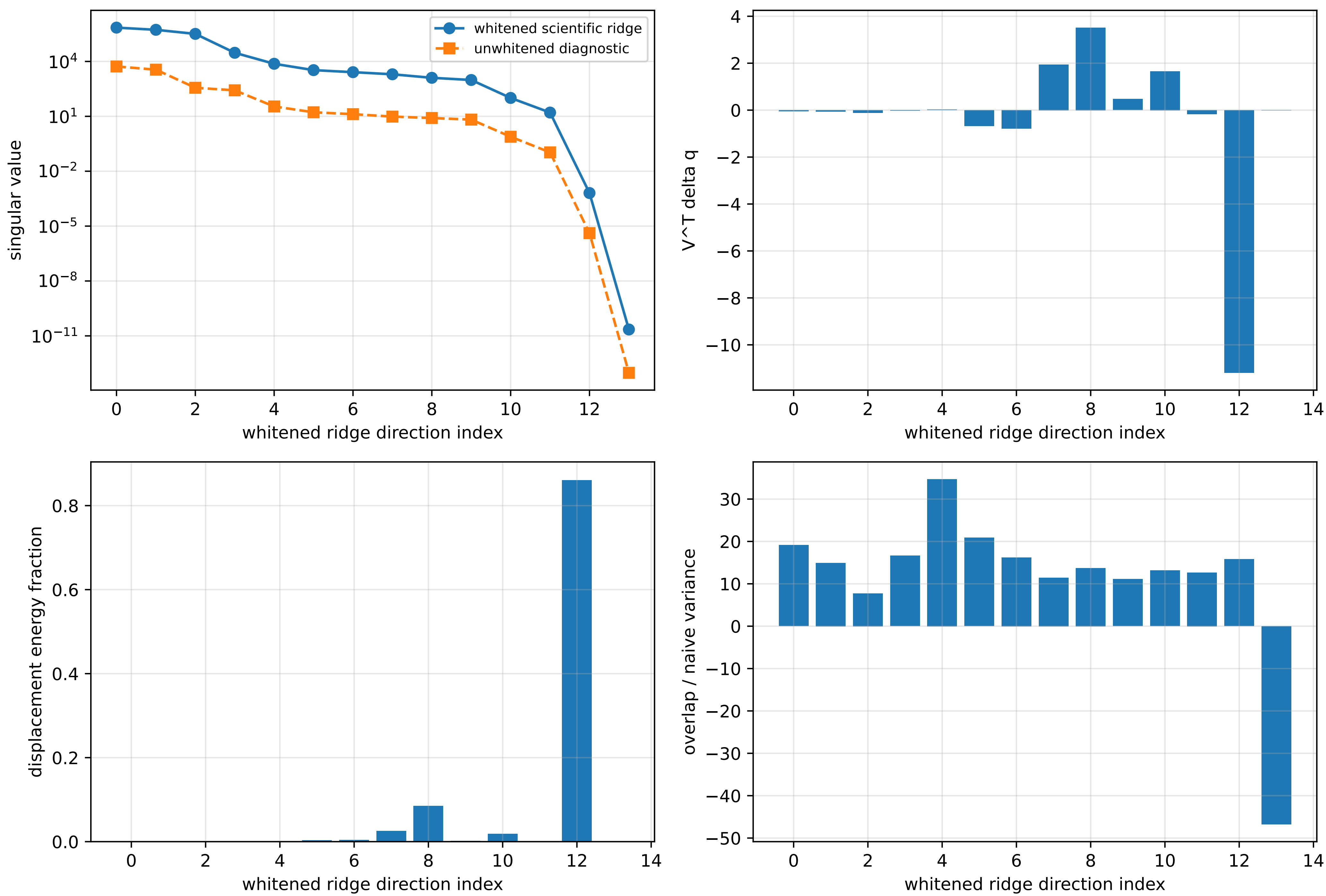

sweep__sg_window_covariance__000: measured sensor comparison
gyro
specific force
measured
predicted
x [rad/s]
y [rad/s]
z [rad/s]
x [m/s^2]
y [m/s^2]
z [m/s^2]
time [s]
0.10
0.05
0.00
−0.05
−0.10
0.15
−0.15
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
1.00
0.75
0.50
0.25
−0.25
−0.50
−0.75
9.85
9.80
9.75
9.70
9.65
9.60
9.55
0
2
4
6
8

sweep__sg_window_covariance__000: nominal -> estimated parameters

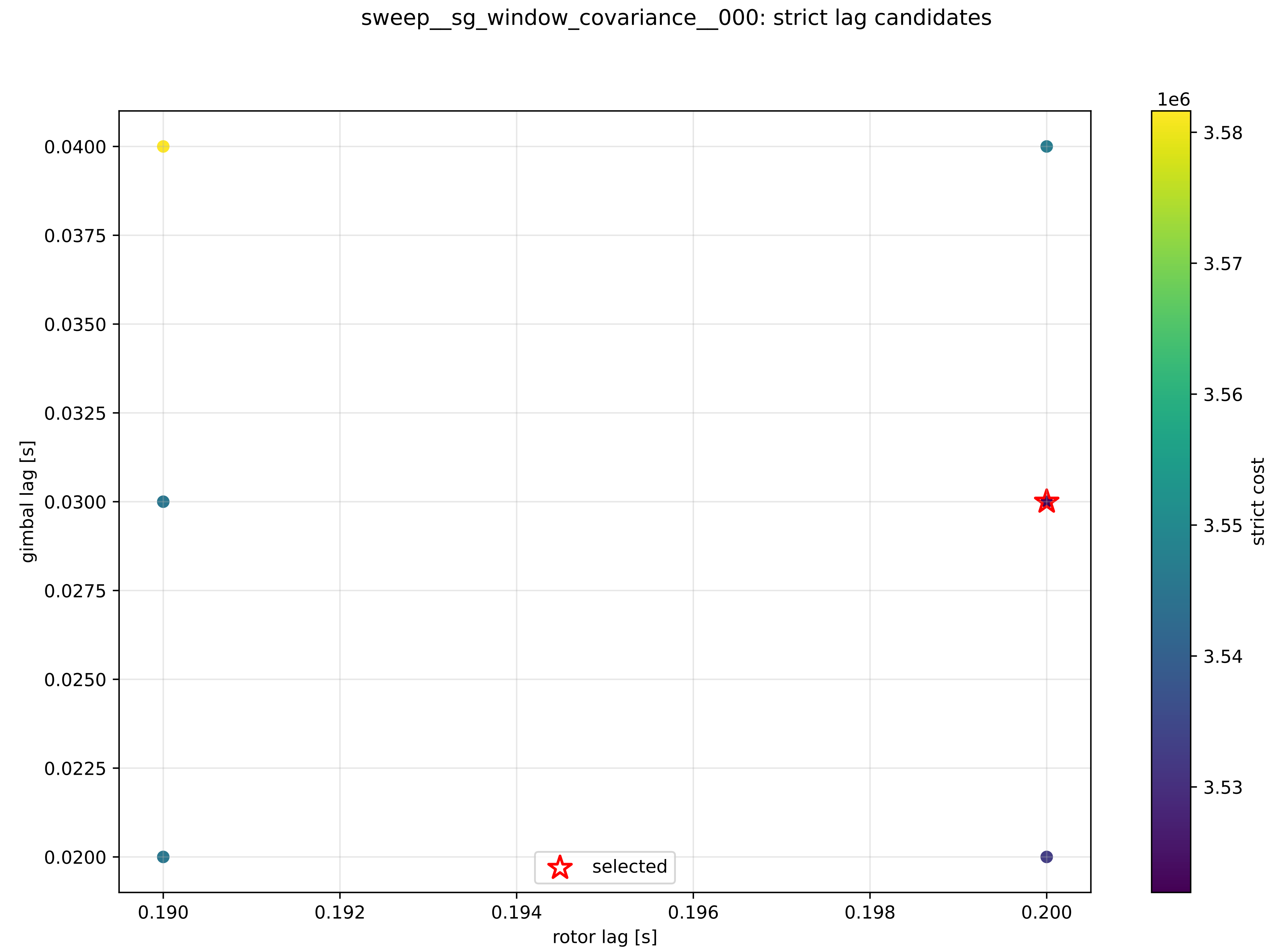
```
Case: sweep__sg_window_covariance__000
status: completed (all stages completed)

mass [kg]       nominal  2.35159759   estimated  5.53357406   delta  3.18197647

inertia [kg m^2] (nominal -> estimated; delta)
  [ 6.5000061e-02 -7.2789925e-07  1.9015080e-05] -> [0.4615689 0.0724134 0.1983013]   d=[0.3965688 0.0724141 0.1982823]
  [-7.2789925e-07  6.4952656e-02  5.9167305e-08] -> [ 0.0724134  0.4666753 -0.0050462]   d=[ 0.0724141  0.4017226 -0.0050463]
  [1.9015080e-05 5.9167305e-08 1.2899211e-01] -> [ 0.1983013 -0.0050462  0.6961332]   d=[ 0.1982823 -0.0050463  0.5671411]

CoG [m]       [-2.0247086e-03 -3.0526579e-05  9.5097496e-03] -> [-0.0246673 -0.0584786  0.0211272]   d=[-0.0226426 -0.058448
rotor force effectiveness
  [1. 1. 1. 1.] -> [2.4975379 2.4496907 1.1013051 1.6301678]   d=[1.4975379 1.4496907 0.1013051 0.6301678]

rotor lag [s]  0.2
gimbal lag [s]  0.0300003291
```


sweep__sg_window_covariance__000: covariance weighting
Sigma_z eigenvalue
10^{-3}
10^{-4}
10^{-5}
10^{-6}
mode 0
mode 1
mode 2
mode 3
mode 4
mode 5
whitening gain
10^{3}
10^{2}
Mahalanobis contribution
20000
15000
10000
5000
0
0
2
4
6
8
time [s]

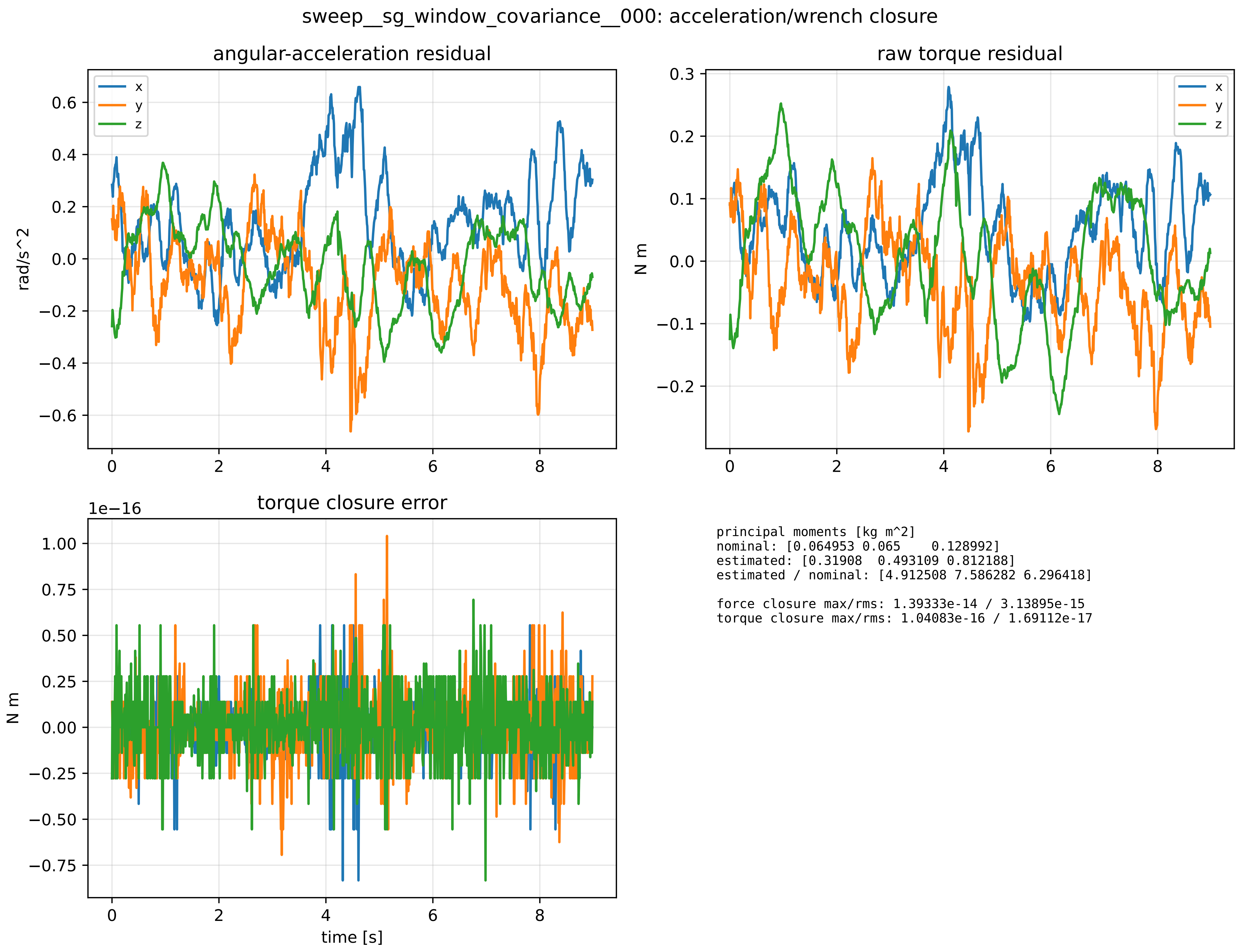
sweep__sg_window_covariance__000: parameter ridge



exact scale gauge: ||J v_scale||=6.40881e-11, rank=13, nullity=1



boundary flags: rotor lower=False upper=True; gimbal lower=False upper=False


sweep__sg_window_covariance__000: acceleration/wrench closure
angular-acceleration residual
rad/s^2
x
y
z
raw torque residual
N m
x
y
z
torque closure error
1e−16
N m
time [s]
principal moments [kg m^2]
nominal: [0.064953 0.065   0.128992]
estimated: [0.31908 0.493109 0.812188]
estimated / nominal: [4.912508 7.586282 6.296418]
force closure max/rms: 1.39333e-14 / 3.13895e-15
torque closure max/rms: 1.04083e-16 / 1.69112e-17